

Pluto 移植指南

移植前提：安装虚拟机 Ubuntu 或 Ubuntu 系统，并安装了 Vivado2023.2 开发套件。

可以正常访问 github、google 网站。

说明：本教程仅说明如何移植 ADI 官方 Pluto 固件工程，不做任何修改。对于需要修改适配后的固件工程，请自行研究。

安装依赖：

```
sudo apt-get install git build-essential fakeroot libncurses5-dev libssl-dev ccache
```

```
sudo apt-get install dfu-util u-boot-tools device-tree-compiler mtools
```

```
sudo apt-get install bc python cpio zip unzip rsync file wget
```

```
sudo apt-get install libtinfo5 device-tree-compiler bison flex u-boot-tools
```

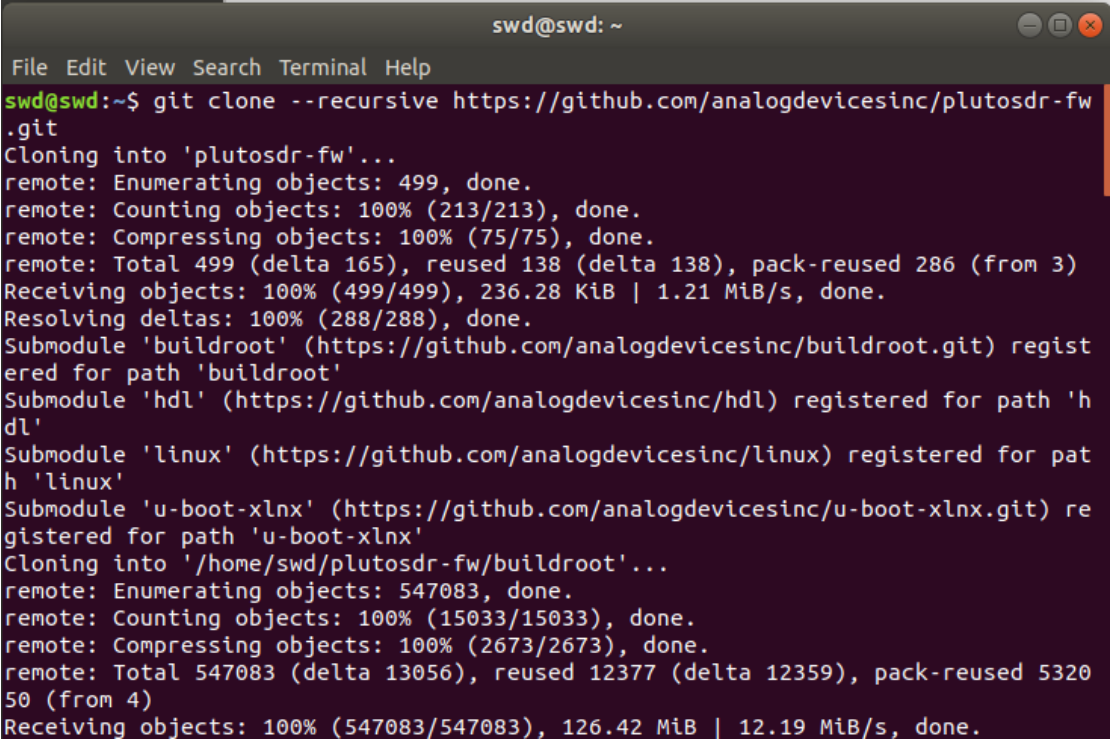
```
sudo apt-get purge gcc-arm-linux-gnueabi
```

```
sudo apt-get install libmpc-dev
```

```
sudo apt-get remove libfdt-dev
```

下载固件：

```
git clone --recursive https://github.com/analogdevicesinc/plutosdr-fw.git
```



```
swd@swd: ~  
File Edit View Search Terminal Help  
swd@swd:~$ git clone --recursive https://github.com/analogdevicesinc/plutosdr-fw  
.git  
Cloning into 'plutosdr-fw'...  
remote: Enumerating objects: 499, done.  
remote: Counting objects: 100% (213/213), done.  
remote: Compressing objects: 100% (75/75), done.  
remote: Total 499 (delta 165), reused 138 (delta 138), pack-reused 286 (from 3)  
Receiving objects: 100% (499/499), 236.28 KiB | 1.21 MiB/s, done.  
Resolving deltas: 100% (288/288), done.  
Submodule 'buildroot' (https://github.com/analogdevicesinc/buildroot.git) regist  
ered for path 'buildroot'  
Submodule 'hdl' (https://github.com/analogdevicesinc/hdl) registered for path 'h  
dl'  
Submodule 'linux' (https://github.com/analogdevicesinc/linux) registered for pat  
h 'linux'  
Submodule 'u-boot-xlnx' (https://github.com/analogdevicesinc/u-boot-xlnx.git) re  
gistered for path 'u-boot-xlnx'  
Cloning into '/home/swd/plutosdr-fw/buildroot'...  
remote: Enumerating objects: 547083, done.  
remote: Counting objects: 100% (15033/15033), done.  
remote: Compressing objects: 100% (2673/2673), done.  
remote: Total 547083 (delta 13056), reused 12377 (delta 12359), pack-reused 5320  
50 (from 4)  
Receiving objects: 100% (547083/547083), 126.42 MiB | 12.19 MiB/s, done.
```

下载完成后，可以看到会显示 hash 值，可以和 github 官网 v0.39 版本 hash 值比对，完全一致。

```
swd@swd: ~  
File Edit View Search Terminal Help  
Resolving deltas: 100% (876498/876498), done.  
Cloning into '/home/swd/plutosdr-fw/linux'...  
remote: Enumerating objects: 12676964, done.  
remote: Counting objects: 100% (10866/10866), done.  
remote: Compressing objects: 100% (1967/1967), done.  
remote: Total 12676964 (delta 10292), reused 8899 (delta 8899), pack-reused 1266  
6098 (from 2)  
Receiving objects: 100% (12676964/12676964), 3.95 GiB | 13.00 MiB/s, done.  
Resolving deltas: 100% (10667458/10667458), done.  
Submodule path 'buildroot': checked out 'e783aadcebad3413b3d60fcfe98a25eb395d83  
9'  
Submodule path 'hdl': checked out '065c8f186ef87ff049d279ed5859ee8d9d91808'  
remote: Enumerating objects: 2, done.  
remote: Counting objects: 100% (1/1), done.  
remote: Total 2 (delta 1), reused 1 (delta 1), pack-reused 1 (from 1)  
Unpacking objects: 100% (2/2), done.  
From https://github.com/analogdevicesinc/linux  
* branch f3da30df60047dc5a0b8fa8c640be774e0f784d9 -> FETCH  
_HEAD  
Submodule path 'linux': checked out 'f3da30df60047dc5a0b8fa8c640be774e0f784d9'  
Submodule path 'u-boot-xlnx': checked out '90401ce9ce029e5563f4dface63914d42badf  
5bc'  
Submodule path 'linux': checked out 'f3da30df60047dc5a0b8fa8c640be774e0f784d9'  
swd@swd:~$
```

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 Lewluu ci: Implement sync to internal resources 8456a77 · 2 months ago 159 Commits

.github/workflows	ci: Implement sync to internal resources	2 months ago
CI	CI/status_rule: Add edge-case	2 years ago
buildroot @ e783aad	PlutoSDR: Prepare for v0.39 release	2 years ago
hdl @ 065c8f1	PlutoSDR: Prepare for v0.39 release	2 years ago
linux @ f3da30d	PlutoSDR: Prepare for v0.39 release	2 years ago
scripts	scripts: Add run-xsdb.tcl	3 years ago
u-boot-xlnx @ 90401ce	PlutoSDR: Prepare for v0.35 release	4 years ago
.gitignore	.gitignore: Update gitignores with auto-generated files	3 years ago
.gitmodules	build: remove default branch for buildroot submodule	7 years ago
LICENSE.md	LICENSE.md: Update with some wording tweaks,	8 years ago
Makefile	PlutoSDR: Prepare for v0.39 release	2 years ago
README.md	Makefile: Switch to buildroot Linaro toolchain	3 years ago
download_and_test.sh	download_and_test: put the pluto into dfu mode, download ...	5 years ago
setup_env.sh	setup_env: set the missing environmental/shell vars that nee...	5 years ago


```
swd@swd: ~/plutosdr-fw
legal-info/README
legal-info/host-manifest.csv
legal-info/host-sources/
legal-info/host-sources/acl-2.3.1/
legal-info/host-sources/acl-2.3.1/series
legal-info/host-sources/acl-2.3.1/acl-2.3.1.tar.xz
legal-info/host-sources/flex-2.0.4/0001-build-with-old-gcc-versions.patch
legal-info/host-sources/flex-2.0.4/
legal-info/host-sources/flex-2.0.4/flex-build-make-it-possible-to-disable-the-build-of-the-f.patch
legal-info/host-sources/flex-2.0.4/flex-2.0.4.tar.gz
legal-info/host-sources/libconfuse-3.3/
legal-info/host-sources/libconfuse-3.3/series
legal-info/host-sources/libconfuse-3.3/confuse-3.3.tar.xz
legal-info/host-sources/libconfuse-3.3/0001-Fix-163-undefined-username-used-with-getpwnam.patch
legal-info/host-sources/cnake-3.22.3/
legal-info/host-sources/cnake-3.22.3/0001-rename-cnake-rootfile.patch
legal-info/host-sources/cnake-3.22.3/series
legal-info/host-sources/cnake-3.22.3/cnake-3.22.3.tar.gz
legal-info/host-sources/bliss-3.8.2/
legal-info/host-sources/bliss-3.8.2/bliss-3.8.2.tar.xz
legal-info/host-sources/gettext-tiny-0.3.2/
legal-info/host-sources/gettext-tiny-0.3.2/gettext-tiny-0.3.2.tar.gz
legal-info/host-sources/gettext-tiny-0.3.2/gettext-0.20.1.tar.xz
legal-info/host-sources/lzip-1.23/
legal-info/host-sources/lzip-1.23/lzip-1.23.tar.gz
legal-info/host-sources/autoconf-2.71/
legal-info/host-sources/autoconf-2.71/series
legal-info/host-sources/autoconf-2.71/autoconf-2.71.tar.xz
legal-info/host-sources/autoconf-2.71/0001-dont-add-dirty-to-version.patch
legal-info/host-sources/patchelf-0.13/
legal-info/host-sources/patchelf-0.13/0001-Add-option-to-make-the-rpath-relative-under-a-specified-path
legal-info/host-sources/patchelf-0.13/patchelf-0.13.tar.bz2
legal-info/host-sources/patchelf-0.13/series
legal-info/host-sources/m4-1.4.19/
legal-info/host-sources/m4-1.4.19/m4-1.4.19.tar.xz
legal-info/host-sources/genimage-16/
legal-info/host-sources/genimage-16/genimage-16.tar.xz
legal-info/host-sources/fakeroot-1.30.1/
legal-info/host-sources/fakeroot-1.30.1/fakeroot-1.30.1.orig.tar.gz
legal-info/host-sources/libtool-2.4.6/
legal-info/host-sources/libtool-2.4.6/libtool-2.4.6.tar.xz
legal-info/host-sources/dosfstools-4.2/
legal-info/host-sources/dosfstools-4.2/dosfstools-4.2.tar.gz
legal-info/host-sources/automake-1.16.5/
legal-info/host-sources/automake-1.16.5/series
legal-info/host-sources/automake-1.16.5/0001-nanon.patch
legal-info/host-sources/automake-1.16.5/automake-1.16.5.tar.xz
legal-info/host-sources/attr-2.5.1/
legal-info/host-sources/attr-2.5.1/0001-build-with-older-GCCs.patch
legal-info/host-sources/attr-2.5.1/attr-2.5.1.tar.xz
legal-info/host-sources/mtools-4.0.38/
legal-info/host-sources/mtools-4.0.38/mtools-4.0.38.tar.lz
legal-info/host-sources/pkgconf-1.6.3/
legal-info/host-sources/pkgconf-1.6.3/0001-Only-prefix-with-the-sysroot-a-subset-of-variables.patch
legal-info/host-sources/pkgconf-1.6.3/pkgconf-1.6.3.tar.xz
legal-info/host-sources/pkgconf-1.6.3/series
legal-info/host-sources/pkgconf-1.6.3/0002-Revert-main-assume-modversion-instead-of-version-1f-o.patch
legal-info/host-sources/uboot-tools-2021.07/
legal-info/host-sources/uboot-tools-2021.07/0001-drop-config-from-tools.patch
legal-info/host-sources/uboot-tools-2021.07/uboot-tools-2021.07/series
legal-info/host-sources/uboot-tools-2021.07/0003-tools-Makefile-Fix-C-ld-flags-with-CROSS_BUILD_TOOLS.patch
legal-info/host-sources/uboot-tools-2021.07/0002-tools-only-in-no-dot-config-targets.patch
legal-info/host-sources/uboot-tools-2021.07/u-boot-2021.07.tar.bz2
rm linux/arch/arm/boot/dts/zyng-pluto-sdr-rev.dtb linux/arch/arm/boot/dts/zyng-pluto-sdr.dtb linux/arch/arm/boot/dts/zyng-pluto-sdr-revb.dtb
swd@swd:~/plutosdr-fw$
```

输入 `ls -AGhl build` 指令，可以看到列出的构建结果。

```
swd@swd: ~/plutosdr-fw
File Edit View Search Terminal Help
legal-info/host-sources/patchelf-0.13/
legal-info/host-sources/patchelf-0.13/0001-Add-option-to-make-the-rpath-relative-under-a-specified-path
legal-info/host-sources/patchelf-0.13/patchelf-0.13.tar.bz2
legal-info/host-sources/patchelf-0.13/series
legal-info/host-sources/m4-1.4.19/
legal-info/host-sources/m4-1.4.19/m4-1.4.19.tar.xz
legal-info/host-sources/genimage-16/
legal-info/host-sources/genimage-16/genimage-16.tar.xz
legal-info/host-sources/fakeroot-1.30.1/
legal-info/host-sources/fakeroot-1.30.1/fakeroot-1.30.1.orig.tar.gz
legal-info/host-sources/libtool-2.4.6/
legal-info/host-sources/libtool-2.4.6/libtool-2.4.6.tar.xz
legal-info/host-sources/dosfstools-4.2/
legal-info/host-sources/dosfstools-4.2/dosfstools-4.2.tar.gz
legal-info/host-sources/automake-1.16.5/
legal-info/host-sources/automake-1.16.5/series
legal-info/host-sources/automake-1.16.5/0001-nanon.patch
legal-info/host-sources/automake-1.16.5/automake-1.16.5.tar.xz
legal-info/host-sources/attr-2.5.1/
legal-info/host-sources/attr-2.5.1/0001-build-with-older-GCCs.patch
legal-info/host-sources/attr-2.5.1/attr-2.5.1.tar.xz
legal-info/host-sources/mtools-4.0.38/
legal-info/host-sources/mtools-4.0.38/mtools-4.0.38.tar.lz
legal-info/host-sources/pkgconf-1.6.3/
legal-info/host-sources/pkgconf-1.6.3/0001-Only-prefix-with-the-sysroot-a-subset-of-variables.patch
legal-info/host-sources/pkgconf-1.6.3/pkgconf-1.6.3.tar.xz
legal-info/host-sources/pkgconf-1.6.3/series
legal-info/host-sources/pkgconf-1.6.3/0002-Revert-main-assume-modversion-instead-of-version-1f-o.patch
legal-info/host-sources/uboot-tools-2021.07/
legal-info/host-sources/uboot-tools-2021.07/0001-drop-config-from-tools.patch
legal-info/host-sources/uboot-tools-2021.07/uboot-tools-2021.07/series
legal-info/host-sources/uboot-tools-2021.07/0003-tools-Makefile-Fix-C-ld-flags-with-CROSS_BUILD_TOOLS.patch
legal-info/host-sources/uboot-tools-2021.07/0002-tools-only-in-no-dot-config-targets.patch
legal-info/host-sources/uboot-tools-2021.07/u-boot-2021.07.tar.bz2
rm linux/arch/arm/boot/dts/zyng-pluto-sdr-rev.dtb linux/arch/arm/boot/dts/zyng-pluto-sdr.dtb linux/arch/arm/boot/dts/zyng-pluto-sdr-revb.dtb
swd@swd:~/plutosdr-fw$ ls -AGhl build
total 537M
-rw-rw-r-- 1 swd 69 1月 22 21:17 boot.bif
-rw-rw-r-- 1 swd 444K 1月 22 21:17 boot.bin
-rw-rw-r-- 1 swd 444K 1月 22 21:17 boot.dfu
-rw-rw-r-- 1 swd 573K 1月 22 21:17 boot.frm
-rw-rw-r-- 1 swd 460M 1月 22 21:17 legal-info-v0.39-1-g8456.tar.gz
-rw-rw-r-- 1 swd 637K 1月 22 21:01 LICENSE.html
-rw-rw-r-- 1 swd 12M 1月 22 21:17 pluto.dfu
-rw-rw-r-- 1 swd 12M 1月 22 21:17 pluto.frm
-rw-rw-r-- 1 swd 33 1月 22 21:17 pluto.frm.md5
-rw-rw-r-- 1 swd 12M 1月 22 21:17 pluto.itb
-rw-rw-r-- 1 swd 23M 1月 22 21:17 plutosdr-fw-v0.39-1-g8456.zip
-rw-rw-r-- 1 swd 596K 1月 22 21:17 plutosdr-ntag-bootstrap-v0.39-1-g8456.zip
-rw-rw-r-- 1 swd 441K 1月 22 21:16 ps7_init.c
-rw-rw-r-- 1 swd 442K 1月 22 21:16 ps7_init_gpl.c
-rw-rw-r-- 1 swd 4.2K 1月 22 21:16 ps7_init_gpl.h
-rw-rw-r-- 1 swd 3.6K 1月 22 21:16 ps7_init.h
-rw-rw-r-- 1 swd 2.4M 1月 22 21:16 ps7_init.html
-rw-rw-r-- 1 swd 31K 1月 22 21:16 ps7_init.tcl
-rw-rw-r-- 1 swd 6.4M 1月 22 21:07 rootfs.cpio.gz
drwxrwxr-x 6 swd 4.0K 1月 22 21:17 sdk
-rw-rw-r-- 1 swd 941K 1月 22 21:16 system_top.bit
-rw-rw-r-- 1 swd 734K 1月 22 21:16 system_top.xsa
-rwxrwxr-x 1 swd 439K 1月 22 21:17 u-boot.elf
-rw-rw-r-- 1 swd 128K 1月 22 21:17 uboot-env.bin
-rw-rw-r-- 1 swd 129K 1月 22 21:17 uboot-env.dfu
-rw-rw-r-- 1 swd 7.6K 1月 22 21:17 uboot-env.txt
drwxrwxr-x 1 swd 4.5M 1月 22 20:59 zImage
-rw-rw-r-- 1 swd 22K 1月 22 21:07 zynq-pluto-sdr.dtb
-rw-rw-r-- 1 swd 22K 1月 22 21:07 zynq-pluto-sdr-revb.dtb
-rw-rw-r-- 1 swd 24K 1月 22 21:08 zynq-pluto-sdr-rev.dtb
swd@swd:~/plutosdr-fw$
```

在《plotusdr_fw/build》文件夹内，可以找到构建结果。

